CO-2 ASSESSMENT TOOL-2

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Course: Machine Learning

Course Code: ITA0612

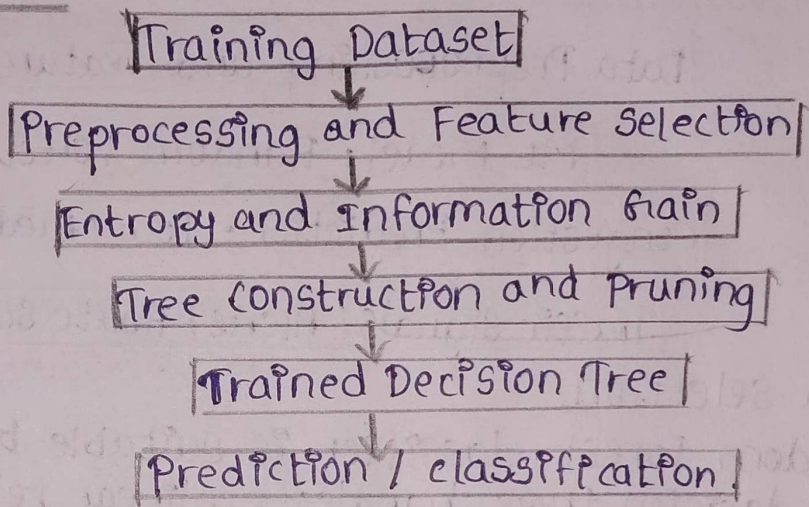
Slot: A

1. With the help of neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Explain how overfitting can be controlled in decision tree models.

Ans:

A Decision Tree is a supervised machine learning algorithm that uses a tree-like structure to classify data or predict outcomes.

Architecture Diagram:



Feature selection:

* Entropy: Measures the impurity or uncertainty in a dataset.

$$H(S) = -\sum p_i \log_2 p_i$$

* Information Gain: Measures the reduction in entropy after splitting the dataset using a feature.

$$IG = H(S) - H(S|A)$$

controlling overfitting:

Use pruning, limit tree depth, set a minimum number of samples per leaf and validate the tree on unseen data.

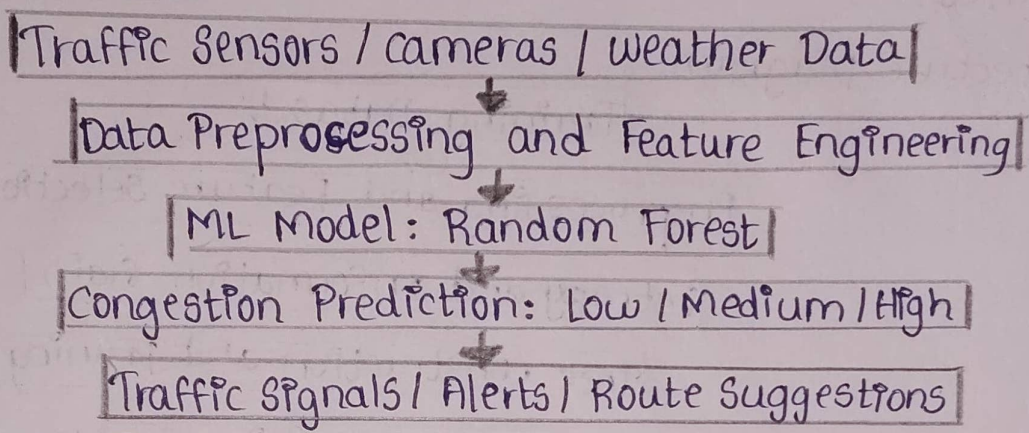
Result:

A decision tree predicts outcomes using selected features and controls overfitting through pruning.

2. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated and deployed for real-time decision-making.

Sol:

System Architecture:



Algorithm Selection:

Random Forest classifier is suitable because it combines multiples decision trees, handles nonlinear relationships, reduces overfitting compared with a single unrestricted tree, and works with different types of features.

Preprocessing and Feature Engineering:

1. Remove duplicate records and handle missing values.
2. Encode categorical features such as weather and road type.
3. Normalize numerical values if required by the chosen model.
4. Extract hour, day of week, vehicle count per minute and congestion history.

Training, Validation and Deployment:

1. Split historical data into training, validation, and test sets, preferably in chronological order.

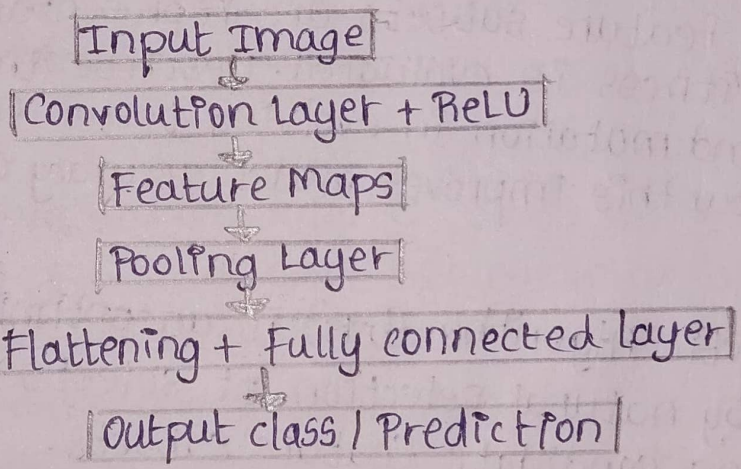
2. Train the Random Forest using the training Data.
3. Tune parameters and evaluate using the validation set; test final performance on unseen data.
4. Deploy the model on a traffic-control server or edge device.
5. Continuously receive sensor data and predict congestion in real time.

3. In the context of Convolutional Neural Networks (CNNs), explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.

Sol:

A CNN is a deep learning model mainly used for image recognition and classification.

CNN Architecture:



Layers and their Functions:

1. Convolutional layer: Applies filters to the input image to detect features such as edges, textures, and shapes.
2. Activation layer (ReLU): Introduces nonlinearity by converting negative values to zero.

3. Pooling Layer: Reduces feature-map dimensions while retaining important information.

4. Flattening Layer: converts feature maps into a one-dimensional vector.

5. Fully connected layer: Uses extracted features to classify the image.

6. Output layer : Produces class probabilities, often using Softmax for multiclass classification.

Factors affecting performance: Training dataset size, image quality, network depth, filter size, learning rate, batch size and overfitting.

Result: CNNs automatically learn image features and classify images effectively through convolution, pooling and weight optimization.

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4. Consider the application of Genetic Algorithms for feature selection in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes, and how fitness is evaluated. Describe the role of selection, crossover and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity.

Ans:

A Genetic Algorithm (GA) is an optimization technique inspired by natural selection. It selects useful features by evolving candidate feature subsets.

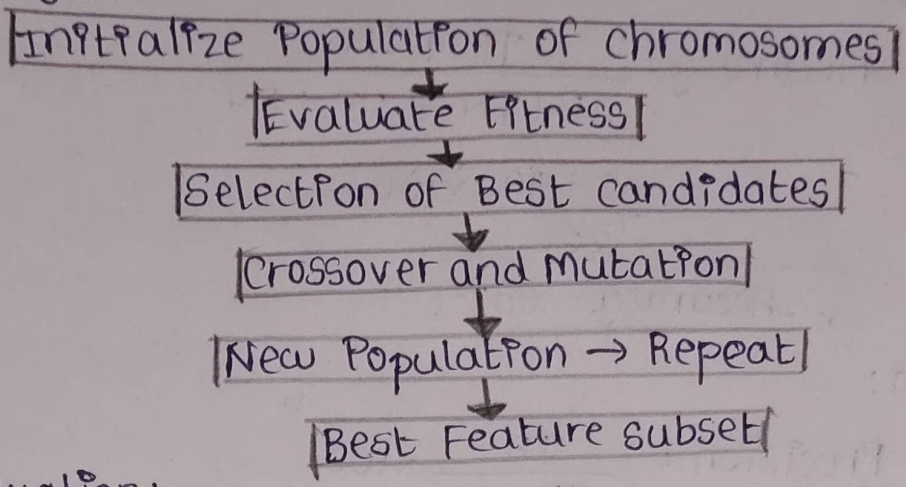
Chromosome Representation:

Suppose there are four features: Age, Income, Credit Score, and Employment.

Each feature is represented by one bit:

- 1011 means Age, Credit Score, and Employment are Selected.
- 0100 means only Income is selected.

Working Diagram:



Fitness Evaluation:

A machine learning classifier is trained using the selected features. Fitness can combine classification performance and feature count:

$$\text{Fitness} = \text{Accuracy} - \lambda\left(\frac{K}{n}\right)$$

Genetic Operation:

- Selection: Chooses chromosomes with better fitness.
- Crossover: Combines parts of two parent chromosomes to generate new feature subsets.
- Mutation: Randomly flips bits, introducing diversity and exploring new subsets.

5. In the context of classification problems, explain how different evaluation metrics such as accuracy, precision, recall, F1-score and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.

Given:

	Predicted Positive	Predicted Negative
Actual Positive	TP = 80	FN = 20
Actual Negative	FP = 10	TN = 90

Total observations: $80 + 20 + 10 + 90 = 200$

1. Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{80 + 90}{200} \times 100$$

$\therefore$ Accuracy = 85%.

2. Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

$$= \frac{80}{80 + 10} \times 100$$

$\therefore$ Precision = 88.89%.

3. Recall (Sensitivity):

$$\text{Recall} = \frac{TP}{TP + FN} \times 100$$

$$= \frac{80}{80 + 20} \times 100$$

$\therefore$ Recall = 80%.

4. F1-Score:

$$F1 = \frac{2TP}{2TP + FP + FN} \times 100$$

$$= \frac{160}{160 + 10 + 20} \times 100$$

$\therefore$ F1 = 84.21%.

Result:

No single metric is sufficient for every classification task. The appropriate metric depends on the cost of false positives and false negatives.